Program-8 Documentation

Title: Create Kubernetes Pods Using YAML Descriptors

1. Minikube setup and status check

```
Terminal
1RV24MC061-leelanjali-p@Radha-Leelanjali:~$ mkdir Program8
1RV24MC061-leelanjali-p@Radha-Leelanjali:~$ cd Program8
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program8$ minikube start
😄  minikube v1.37.0 on Ubuntu 24.04
✨  Using the docker driver based on existing profile
👍  Starting "minikube" primary control-plane node in "minikube" cluster
🚜  Pulling base image v0.0.48 ...
🔄  Restarting existing docker container for "minikube" ...
🐳  Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔎  Verifying Kubernetes components...
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟  Enabled addons: storage-provisioner, default-storageclass

❗  /usr/local/bin/kubectl is version 1.31.0, which may have incompatibilities with Kubernetes 1
.34.0.
    ▪ Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
🏄  Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program8$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program8$ nano server.js
```

Minikube is used here to run a single-node Kubernetes cluster locally.

Command	Purpose	Output/Notes
mkdir Program8	Creates a new directory for the project.	
cd Program8	Changes the current directory to the project folder.	
minikube start	Starts the Minikube cluster. This command pulls necessary images, starts the control plane node, and configures kubectl.	Displays progress: "Starting 'minikube' primary control-plane node...", "Done! kubectl is now configured to use 'minikube' cluster..."
minikube status	Checks the status of the Minikube cluster components.	Confirms all key components (Control Plane, host, kubelet, apiserver, kubeconfig) are Running or Configured.

2. Application and Configuration Files

```
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program8$ nano server.js
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program8$ nano nodejs-configmap.yaml
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program8$ nano nodejs-pod.yaml
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program8$ kubectl get pods
No resources found in default namespace.
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program8$ kubectl apply -f nodejs-configmap.yaml
configmap/nodejs-app-config unchanged
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program8$ kubectl apply -f nodejs-pod.yaml
pod/program-8 created
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program8$ kubectl get pods
NAME        READY   STATUS    RESTARTS   AGE
program-8   1/1     Running   0          3s
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program8$ kubectl port-forward pod/program-8 3000:300
0
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000
```

The screenshots imply the creation of three essential files: the application, a ConfigMap, and the Pod definition.

Command	Purpose
nano server.js	Edits or creates the Node.js application file (server.js) that will run inside the container. This application is responsible for the output: "Hello from Node.js running inside Kubernetes!".
nano nodejs-configmap.yaml	Edits or creates the ConfigMap YAML file. This file defines configuration data (e.g., environment variables, settings) to be injected into the Pod.
nano nodejs-pod.yaml	Edits or creates the Pod Definition YAML file. This file describes the desired state of the Pod, including the container image, ports, and how to use the ConfigMap.

3. Kubernetes Deployment

These steps deploy the configuration and application to the Minikube cluster.

Command	Purpose	Output/Notes
kubectl get pods	Initial check to ensure the namespace is empty before deployment.	No resources found in default namespace.
kubectl apply -f nodejs-configmap.yaml	Applies the ConfigMap definition to the cluster.	configmap/nodejs-app-config unchanged (or created)
kubectl apply -f nodejs-pod.yaml	Applies the Pod definition to the cluster, instructing Kubernetes	pod/program-8 created

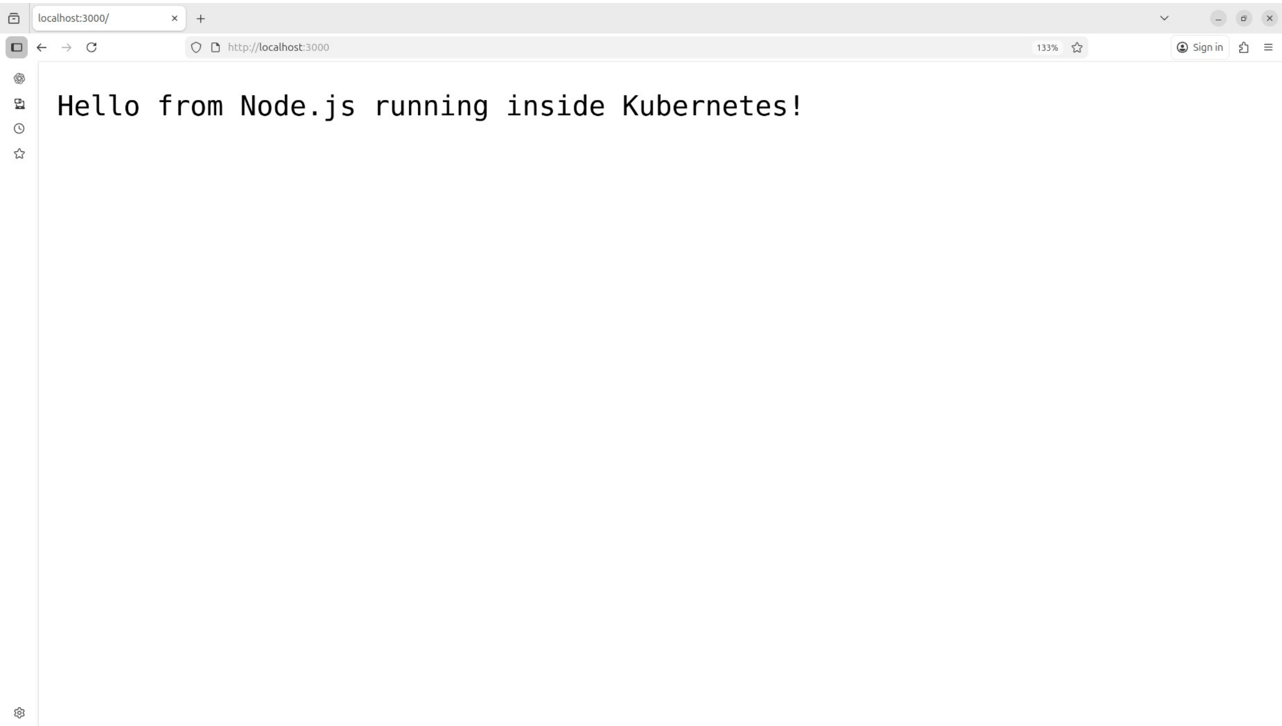
Command	Purpose	Output/Notes
	to create the Pod.	
kubectl get pods	Verifies the Pod creation and status.	Shows the Pod name (program-8), STATUS as Running (1/1 ready), and its AGE.

4. Accessing the Application

The application is running inside the Pod, but it is not directly accessible from the host machine without explicit port mapping.

Command	Purpose	Output/Notes
kubectl port-forward pod/program-8 3000:3000	Sets up local port forwarding. This command routes traffic from a local port (3000 on the host) to a specific port (3000 inside the program-8 Pod).	Forwarding from 127.0.0.1:3000 -> 3000

5. Verification



The final step is to verify the deployed application is running and accessible.

Action	Result
Access http://localhost:3000 in a web browser.	The browser displays: "Hello from Node.js running inside Kubernetes!", confirming successful deployment and access through port forwarding.